

Technical Design: Multi-Source Candidate Data Transformer

Project Objective: The Multi-Source Candidate Data Transformer is an ETL pipeline that consolidates candidate information from multiple sources into a single canonical profile. It resolves duplicate records, inconsistent formats, and conflicting values through extraction, normalization, deduplication, confidence scoring, configurable projection, and schema validation while preserving complete data provenance.

Data Ingestion & Extraction : Candidate data is collected from Recruiter CSV, ATS JSON, GitHub, LinkedIn, Resume, and Recruiter Notes. Dedicated CSV, JSON, and Regex parsers convert structured and unstructured data into a unified candidate object while retaining source information for traceability

Data Normalization : Candidate attributes are standardized before matching. Emails are lowercased, phone numbers converted to E.164, dates formatted to ISO standards, locations normalized to ISO-3166, and skill aliases mapped to canonical names, ensuring consistent data across all sources.

Candidate Matching & Merge Strategy: Duplicate candidates are identified using normalized email and phone numbers. Records are grouped using a Union-Find algorithm, while conflicting values are resolved using source priority (Resume > ATS = GitHub = CSV > LinkedIn > Recruiter Notes). Skills, Experience, and Education are merged and deduplicated.

Confidence Scoring & Provenance : Each field receives a confidence score based on source reliability, agreement across sources, normalization success, and extraction quality. Overall confidence is calculated from field scores, while provenance metadata records the source and extraction method for every attribute.

Runtime Projection & Schema Validation : A runtime-configurable projection layer dynamically maps canonical profiles into custom JSON schemas through configuration files. It supports field remapping, nested extraction, metadata control, and configurable missing-value policies (null, omit, or error). Schema validation verifies data types, required fields, and structural integrity before export.

Edge Cases & Design Considerations : The pipeline handles duplicate candidates, conflicting information, inconsistent formats, duplicate skills, varying phone and date formats, missing optional fields, and partial records while maintaining stability.

